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Fine-Tuning & Distillation

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Validation

29 automated tests

Abstract

This brief presents an adaptation and compression pipeline that implements real fine-tuning, LoRA, and knowledge-distillation mechanics while staying runnable on CPU with a small surrogate model.

Project Background

The point of the project is not to pretend a giant language model was trained locally. It is to show that the engineering logic of adaptation, parameter efficiency, and promotion gating remains valid when the implementation is simplified honestly.

Headline Result

LoRA 91% params; KD recovers 100% of teacher

Scope Overview

Synthetic instruction dataset and hashed feature pipeline. Baseline model, LoRA adapter path, and KD student training. Evaluation tables comparing teacher, adapter, and student outcomes. Promotion gate and scaling notes toward a real LLM stack.

Project Context

A fully-runnable, GPU-free pipeline that implements the core techniques used to adapt and compress large language models - full fine-tuning, LoRA low-rank adaptation, knowledge distillation, an automated eval/promotion gate, and a synthetic-data scaling study - end to end, deterministically, on a CPU in a few minutes.

Executive Summary

Background, objectives, scope, and project impact.

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This brief presents an adaptation and compression pipeline that implements real fine-tuning, LoRA, and knowledge-distillation mechanics while staying runnable on CPU with a small surrogate model.

Background

The point of the project is not to pretend a giant language model was trained locally. It is to show that the engineering logic of adaptation, parameter efficiency, and promotion gating remains valid when the implementation is simplified honestly.

Objectives

- Implement full fine-tuning, LoRA, and KD as real training loops.
- Show parameter-efficiency and teacher-student trade-offs clearly.
- Add an evaluation and promotion gate that mirrors a production workflow.

Scope

- Synthetic instruction dataset and hashed feature pipeline.
- Baseline model, LoRA adapter path, and KD student training.
- Evaluation tables comparing teacher, adapter, and student outcomes.
- Promotion gate and scaling notes toward a real LLM stack.

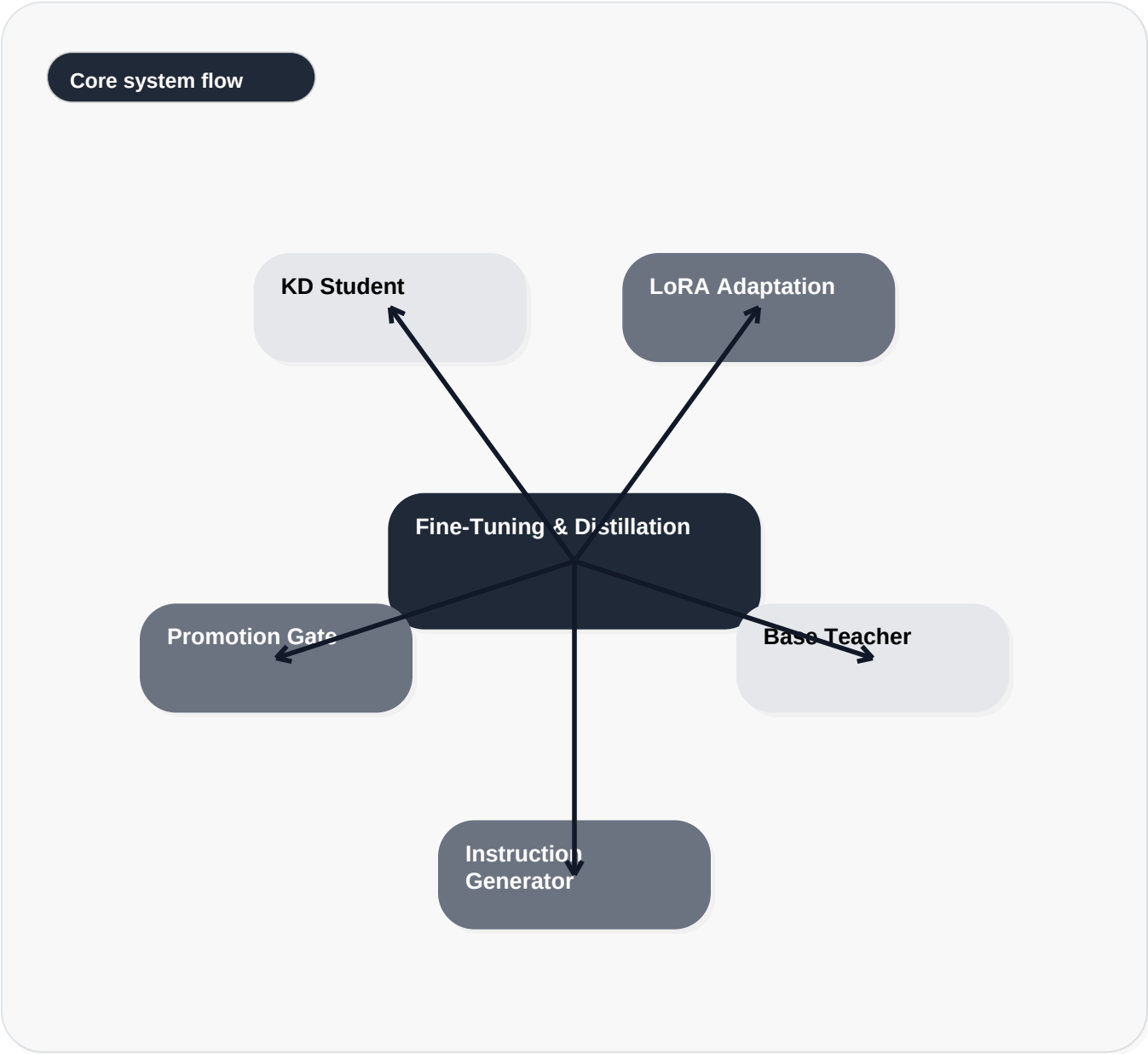
Impact

The final workflow is valuable because it isolates the technique from the hype. It proves what LoRA changes, what distillation recovers, and how a promotion gate should evaluate a challenger before it is accepted.

LoRA 91% params; KD recovers 100% of teacher

Solution Architecture

This flow mirrors a real adaptation stack: a stronger teacher, a parameter-efficient adapter, and a hard promotion gate.



Design Note 1

The surrogate model keeps the story honest while preserving the real optimization mechanics.

Design Note 2

LoRA is presented as a parameterization decision with measurable savings, not a branding label.

Design Note 3

Distillation is evaluated on recovered teacher quality rather than anecdotal fluency claims.

Implementation Process

Delivery phases, engineering approach, and quality checkpoints.

Delivery Narrative

A fully-runnable, GPU-free pipeline that implements the core techniques used to adapt and compress large language models - full fine-tuning, LoRA low-rank adaptation, knowledge distillation, an automated eval/promotion gate, and a synthetic-data scaling study - end to end, deterministically, on a CPU in a few minutes.



Quality Checkpoints

- 29 automated tests validate core behavior and edge cases.
- Benchmark files and screenshots are generated from committed project artifacts.
- The run path stays deterministic so numbers can be reproduced and reviewed directly.

Engineering Principles

- The surrogate model keeps the story honest while preserving the real optimization mechanics.
- LoRA is presented as a parameterization decision with measurable savings, not a branding label.
- Distillation is evaluated on recovered teacher quality rather than anecdotal fluency claims.

Evidence And Results

Test evidence, benchmark interpretation, and screenshot review.

29 passing tests

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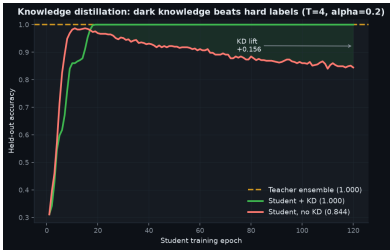
LoRA 91% params; KD recovers 100% of teacher

Benchmark Evidence

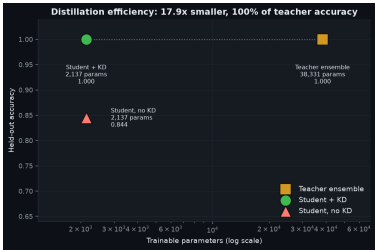
Method	Target-domain accuracy	Trainable params
Full fine-tune	0.934	33,417
LoRA (r=4)	0.934	2,889

Result Interpretation

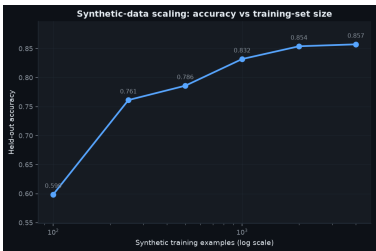
- The charts make the parameter-efficiency story visible because savings and quality can be read together.
- KD recovery matters because compression is only useful if the student preserves enough capability.
- The evaluation gate turns the project from a training experiment into a release decision workflow.



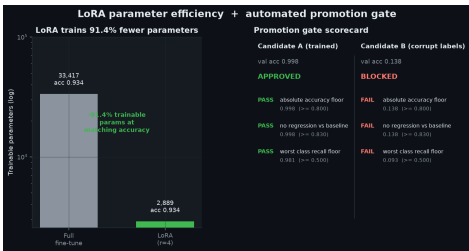
Knowledge distillation: dark knowledge beats hard labels



Distillation efficiency (parameters vs accuracy)



Synthetic-data scaling



LoRA parameter efficiency + promotion-gate scorecard

Risks And Next Steps

Operational considerations, mitigation choices, and forward path.

Risk Register

Misreading the surrogate

Handled by explaining clearly what maps directly to real LLM stacks and what does not.

Quality regression after compression

Managed through teacher-student evaluation and promotion gating.

Technique overclaiming

Avoided by limiting claims to what the measured pipeline actually demonstrates.

Next Steps

- Swap the surrogate model for a transformer implementation when GPU resources are available.
- Add richer task slices so promotion decisions reflect multiple instruction categories.
- Track adapter merge behavior and inference cost under a serving-oriented benchmark.

Closing Note

The project brief is intentionally written as delivered work rather than a transfer note. It documents the business problem, the engineering choices, the measured evidence, and the remaining operational path in a format suitable for portfolio review.

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